

CSE330: NUMERICAL METHODS

ASSIGNMENT 02

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Answer to the Question No - 1

Given. $f(x) = \sin(x) + \cos(x)$; Nodes $\{-\pi, -\frac{\pi}{2}, 0\}$

x	$f(x)$
$-\pi$	-1
$-\frac{\pi}{2}$	-1
0	1
$\frac{\pi}{2}$	1

Node = 3 , Degree, $n = 2$

New Node = 4, New degree = 3

(a)

$n = 2$,

$$P_2(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1)$$

Computing Newton basis,

$$x_0 = -\pi \quad f[x_0] = -1$$

$$f[x_0, x_1] = \frac{-1+1}{-\frac{\pi}{2}+\pi} = 0$$

$$x_1 = -\frac{\pi}{2} \quad f[x_1] = -1$$

$$f[x_0, x_1, x_2] = \frac{\frac{1}{\pi} - 0}{0+\pi} = \frac{1}{\pi^2}$$

$$x_2 = 0 \quad f[x_2] = 1$$

$$f[x_1, x_2] = \frac{1+1}{0+\frac{\pi}{2}} = \frac{4}{\pi}$$

$$f[x_0, x_1, x_2, x_3] = \frac{-\frac{4}{\pi^2} - \frac{4}{\pi^2}}{\frac{\pi}{2} + \pi} = \frac{-16}{3\pi^2}$$

$$x_3 = \frac{\pi}{2} \quad f[x_3] = 1$$

$$f[x_2, x_3] = \frac{1-1}{\frac{\pi}{2}-0} = 0$$

$$f[x_1, x_2, x_3] = \frac{0 - \frac{4}{\pi}}{\frac{\pi}{2} + \frac{\pi}{2}} = -\frac{4}{\pi^2}$$

$\therefore$ basis are, $f[x_0] = -1$, $f[x_0, x_1] = 0$, $f[x_0, x_1, x_2] = \frac{1}{\pi^2}$

Newton Interpolation Polynomial,

$$P_2(x) = -1 + 0(x+\pi) + \frac{1}{\pi^2}(x+\pi)(x+\frac{\pi}{2})$$

$$\therefore P_2(x) = -1 + \frac{1}{\pi^2}(x+\pi)(x+\frac{\pi}{2})$$

(b)

from 'a', $n=2$; Interval is not specified so taking from smallest node to the largest, $[-\pi, 0]$

$$\text{Upper bound} \leq \left| \frac{f^{(2+1)}(\xi)}{(2+1)!} \right|_{\max_{\xi \in (-\pi, 0)}} \left| w_2(x) \right|_{\max_{x \in [-\pi, 0]}}$$

$$\left| \frac{f^{(2+1)}(\xi)}{3!} \right|_{\max_{\xi \in (-\pi, 0)}} = \frac{1}{3!} \left| -\cos(\xi) + \sin(\xi) \right|_{\max_{\xi \in (-\pi, 0)}}$$

$$= \frac{1}{3!} \left| -\cos\left(-\frac{\pi}{4}\right) + \sin\left(-\frac{\pi}{4}\right) \right|$$

$$= \frac{1}{6} | -\sqrt{2} |$$

$$= \frac{\sqrt{2}}{6}$$

$$w_2(x) = (x-x_0)(x-x_1)(x-x_2)$$

$$= (x+\pi)\left(x+\frac{\pi}{2}\right)(x-0)$$

$$= x^3 + x^2 \frac{3\pi}{2} + x \frac{\pi^2}{2}$$

$$\left| w_2(x) \right|_{\max_{x \in [-\pi, 0]}} = \left| w_2(x_2) \right| = 1.491790182$$

$$\therefore \text{Upper bound} \leq \frac{\sqrt{2}}{6} \times 1.491790182$$

$$\therefore \text{Upper bound} \leq 0.3516183179$$

$$f'(x) = \cos x - \sin x$$

$$f''(x) = -\sin x - \cos x$$

$$f'''(x) = -\cos x + \sin x$$

$$\frac{d}{dx}(-\cos x + \sin x) = \sin x + \cos x$$

$$\text{Let, } \sin x + \cos x = 0$$

$$\Rightarrow \tan x = -\cos x$$

$$\Rightarrow \tan x = -1$$

$$\Rightarrow x = -\frac{\pi}{4}$$

(in the range)

$$w'(x) = 3x^2 + 3\pi x + \frac{\pi^2}{2}$$

$$\text{Let, } 3x^2 + 3\pi x + \frac{\pi^2}{2} = 0$$

$$\Rightarrow 6x^2 + 6\pi x + \pi^2 = 0$$

$$\therefore x = \frac{-6\pi \pm \sqrt{(6\pi)^2 - 4 \times 6 \times \pi^2}}{2 \times 6}$$

$$x = -\frac{\pi}{2} \pm \frac{\sqrt{3}}{6} \pi$$

$$\left. \begin{aligned} x_1 &= -\frac{\pi}{2} + \frac{\sqrt{3}}{6} \pi \\ x_2 &= -\frac{\pi}{2} - \frac{\sqrt{3}}{6} \pi \end{aligned} \right\} \text{(in the range)}$$

$$w(\pi) = w\left(-\frac{\pi}{2}\right) = w(0) = 0$$

$$w(x_1) = -1.491790182$$

$$w(x_2) = 1.491790182$$

c)

Adding new node in "a" the basis computation in 'a' using Newton method.

Now, degree, $n=3$

$$P_3(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2)$$

$$P_3(x) = -1 + 0(x+\pi) + \frac{4}{\pi^2}(x+\pi)(x+\frac{\pi}{2}) + (\frac{-16}{3\pi^3})(x+\pi)(x+\frac{\pi}{2})(x-0)$$

$$\therefore P_3(x) = -1 + \frac{4}{\pi^2}(x+\pi)(x+\frac{\pi}{2}) - \frac{16}{3\pi^3}(x+\pi)(x+\frac{\pi}{2})x$$

Answer to the Question No-2

$n=5$, $\therefore j=0,1,2,3,4,5$; and Interval $[-2,3]$, $b=3, a=-2$

Chebyshev points,

$$\varphi_j = \frac{(2j+1)\pi}{2(n+1)}$$

$$\varphi_0 = \frac{\pi}{12}$$

$$\varphi_1 = \frac{\pi}{4}$$

$$\varphi_2 = \frac{5\pi}{12}$$

$$\varphi_3 = \frac{7\pi}{12}$$

$$\varphi_4 = \frac{3\pi}{4}$$

$$\varphi_5 = \frac{11\pi}{12}$$

Chebyshev Nodes,

$$\begin{aligned} x_j &= \left(\frac{b+a}{2}\right) + \left(\frac{b-a}{2}\right) \cos \varphi_j \\ &= \left(\frac{3+(-2)}{2}\right) + \left(\frac{3-(-2)}{2}\right) \cos \varphi_j = \frac{1}{2} + \frac{5}{2} \cos \varphi_j \end{aligned}$$

$$x_0 = \frac{1}{2} + \frac{5}{2} \cos \frac{\pi}{12} = 2.914814566$$

$$x_1 = \frac{1}{2} + \frac{5}{2} \cos \frac{\pi}{4} = 2.267766953$$

$$x_2 = \frac{1}{2} + \frac{5}{2} \cos \frac{5\pi}{12} = 1.147047613$$

$$x_3 = \frac{1}{2} + \frac{5}{2} \cos \frac{7\pi}{12} = -0.1470476128$$

$$x_4 = \frac{1}{2} + \frac{5}{2} \cos \frac{3\pi}{4} = -1.267766953$$

$$x_5 = \frac{1}{2} + \frac{5}{2} \cos \frac{11\pi}{12} = -1.914814566$$

Answer to the Question No-3

Given. $f(x) = \sin(x) - \cos(x)$; Nodes $[0, \pi/2]$

	x	$f(x)$	$f'(x) = \cos(x) + \sin(x)$
Node = 2	0	-1	1
Degree, $n=1$	$\frac{\pi}{2}$	1	1

Hermite node = 4

degree = 3

(a)

Computing lagrange basis,

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - \pi/2}{0 - \pi/2}$$

$$= -\frac{2}{\pi} (x - \frac{\pi}{2})$$

Computing derivatives,

$$l'_0(x) = -\frac{2}{\pi}$$

$$l'_1(x) = \frac{2}{\pi}$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - 0}{\frac{\pi}{2} - 0} = \frac{2}{\pi} x$$

(b)

$$P_{2n+1}(x) = P_{2n+1}(x) = P_3(x) = h_0(x)f(x) + \hat{h}_0(x)f'(x) + h_1(x)f(x) + \hat{h}_1(x)f'(x)$$

Computing hermite basis, $h_k(x) = \{1 - 2(x - x_k)l'_k(x)\} l_k^2(x)$ & $\hat{h}_k(x) = (x - x_k) l_k^2(x)$

$$h_0 = \{1 - 2(x - 0)(-\frac{2}{\pi})\} \{-\frac{2}{\pi}(x - \frac{\pi}{2})\}^2 \quad \hat{h}_0 = (x - 0) \{-\frac{2}{\pi}(x - \frac{\pi}{2})\}^2$$

$$h_1 = \{1 - 2(x - \frac{\pi}{2})\frac{2}{\pi}\} (\frac{2x}{\pi})^2 \quad \hat{h}_1 = (x - \frac{\pi}{2}) (\frac{2x}{\pi})^2$$

$\therefore$ Hermite polynomial,

$$P_3(x) = \{1 - 2x(-\frac{2}{\pi})\} \{-\frac{2}{\pi}(x - \frac{\pi}{2})\}^2 (-1) + x \{-\frac{2}{\pi}(x - \frac{\pi}{2})\}^2 (1)$$

$$+ \{1 - 2(x - \frac{\pi}{2})\frac{2}{\pi}\} (\frac{2x}{\pi})^2 (1) + (x - \frac{\pi}{2}) (\frac{2x}{\pi})^2 (1)$$

Answer to the Question No-4

Given, $f(x) = x \ln(3x) + x^2$; upto 5 significant figure

(a) $x_0 = 2, h = 0.1$

$$D_{h, \text{central}} = \frac{f(x_0+h) - f(x_0-h)}{2h}$$

$$= \frac{f(2+0.1) - f(2-0.1)}{2 \times 0.1}$$

$$= \frac{f(2.1) - f(1.9)}{0.2}$$

$$= \frac{8.2751 - 6.9170}{0.2}$$

$$= \frac{1.3581}{0.2}$$

$$= 6.7905$$

$$f(2.1) = 2.1 \times \ln(3 \times 2.1) + (2.1)^2$$

$$= 2.1 \times 1.8405 + 4.4100$$

$$= 3.8651 + 4.4100$$

$$= 8.2751$$

$$f(1.9) = 1.9 \times \ln(3 \times 1.9) + (1.9)^2$$

$$= 1.9000 \times 1.7405 + 3.6100$$

$$= 3.3070 + 3.6100$$

$$= 6.9170$$

(b) $f'(x) = x \frac{3}{3x} + \ln(3x) + 2x$

$$= 2x + \ln(3x) + 1$$

$f'(2) = 6.7918$ Exact

Truncation Error $\leq \left| -\frac{h^2}{6} f'''(\xi) \right|_{\max}$
 $\xi \in (x_0-h, x_0+h)$

$$= \left| \left(-\frac{h^2}{6} \right) \left(-\frac{1}{\xi^2} \right) \right|_{\max}$$

$$\xi \in (x_0-h, x_0+h)$$

$$= \left| \left(-\frac{0.1^2}{6} \right) \left(-\frac{1}{(1.9)^2} \right) \right|_{\max}$$

$$= 0.0016667 \times 0.27701$$

$$f''(x) = 2 + \frac{3}{3x} = 2 + \frac{1}{x}$$

$$f'''(x) = -\frac{1}{x^2}$$

$$x_0-h = 1.9$$

$$x_0+h = 2.1$$

$\therefore$ Upper bound Truncation Error ≤ 0.00046169